

Lean Canvas: AI-Powered Personalized Learning Agent for Rural Students

■ Problem • Lack of personalized education in rural areas • Limited access to quality teachers • Language and digital barriers	■ Customer Segments • End Users: Rural school students • Buyers: Government schools and NGOs • Early Adopters: Rural schools in Asansol & Paschim Bardhaman	■ Value Proposition • Affordable AI tutor for rural learning • Adaptive & multilingual experience
■ Solution • AI-based adaptive learning system • Voice & regional language support • Offline functionality	■ Channels • Govt & village council partnerships • School tablet pre-installation • Community distribution (WhatsApp/offline)	■ Revenue Streams • Freemium model • Govt funding (B2G) • NGO & CSR grants
■ Cost Structure • Cloud & LLM API costs • AI development (Python frameworks) • Model training (regional dialects) • Infrastructure	■ Key Metrics • Engagement & session duration • Learning improvement scores • Retention rate	■ Unfair Advantage • Offline-first data caching • Optimized for low-memory Android • Deep multilingual rural focus

AI Tutor Pro: Bridging Rural Gaps

A Lightweight, Multilingual AI Learning Agent for Rural Students. Presented by Aditya Sharma, Final Year B.Tech CSE, Roll No. 78, Asansol, West Bengal.

ADITYA SHARMA



Core Challenges in Rural Education

Remote students face critical obstacles including lack of quality tutoring, expert scarcity, language barriers, and device limitations that hinder learning outcomes.

Educational Gap

Rural students in areas like Paschim Bardhaman lack access to high-quality, personalized tutoring services, resulting in a significant educational divide that prevents them from reaching their full potential.

Teacher Scarcity

There is an extremely limited availability of qualified subject matter experts in remote villages, making it difficult for students to receive expert guidance and support for their academic subjects.

Language Barrier

Most available online educational content is provided exclusively in English, which creates a massive and often insurmountable hurdle for regional language speakers trying to access critical materials.

Hardware Limits

A large portion of students rely on older, low-memory Android devices and unstable internet connections, preventing them from accessing modern digital learning platforms or high-bandwidth content.

The Solution

AI Tutor Pro provides instant academic support via any mobile browser. Using LLMs for expert explanations, it features an intuitive, messaging-app style design.



AI Tutor Pro

A web-based AI companion that provides instant, accurate, and reliable academic support.



No Installation

No app installation is required; the platform works directly on any mobile web browser.



Expert Reasoning

Powered by state-of-the-art LLMs to provide expert-level explanations for free.



Intuitive Design

A clean, bubble-chat interface that mimics modern messaging apps for easy use.

Technical Stack Architecture

A summary of the core technology stack used, highlighting the frontend UI, robust Python backend, server deployment, API integration, and security measures.



Frontend Development

Frontend: Uses HTML5, CSS3, and JavaScript with the Fetch API. This combination creates a responsive, modern user interface for the application.



Backend Framework

Backend: Utilizes Python and the Flask framework. This technology choice provides robust application logic and effective API management for the system.



Production Server

Production: The application uses the Gunicorn server. It is deployed and hosted on the Render platform for reliable and production-ready performance.



API Layer

API Layer: Accesses the OpenRouter service. This layer connects to free-tier models like meta-llama/llama-3.3-70b-instruct:free for data processing.



Security Protocols

Security: All API keys are managed using environment variables. The system uses `os.getenv` to keep keys secure and prevent accidental code exposure.

Lean Canvas Highlights



Deployment & Access

24/7 uptime, zero-cost access, and mobile-first design for students.



Live Deployment: Hosted on Render for 24/7 service availability.



Zero Cost: Built with free-tier APIs to ensure free student access.



Mobile First: CSS Flexbox design for small screens and budget tablets.

Rural Impact & Vision

Leveraging native AI language support and NGO partnerships for learning. Starting in Asansol, we bridge the digital divide for underserved rural communities.



Language Support

AI models natively support Hindi and Bengali, removing language barriers to make high-quality digital learning accessible in regional dialects for local students.



Scaling Asansol

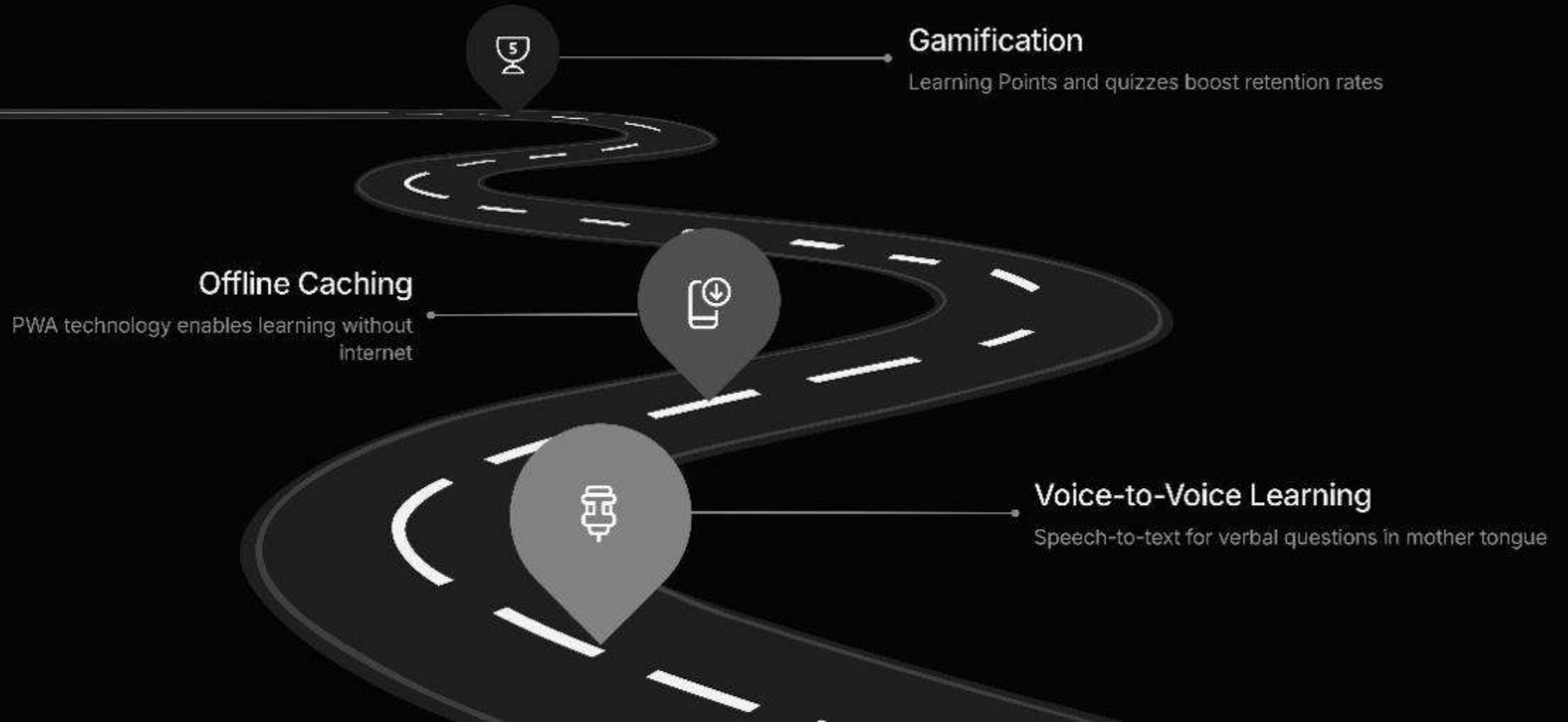
We are rolling out our pilot program across local rural schools in the Asansol region, establishing a scalable foundation for future expansion into more districts.



NGO Partnerships

Our solution is pre-installed on tablets distributed by NGOs and government schemes, ensuring the technology reaches students in need without any extra hurdles.

Future Roadmap



Conclusion

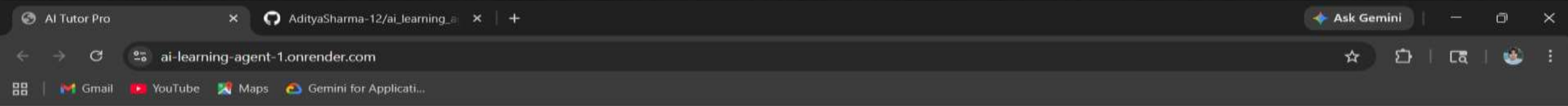


AI Tutor Pro is more than an app—it's a scalable infrastructure for educational equity



Ensure a student in a remote village in West Bengal has the same access to knowledge as a student in a major city

Screenshot



AI Tutor

Type your question here... Send

You: 2+2

AI Tutor: 4

You: hello

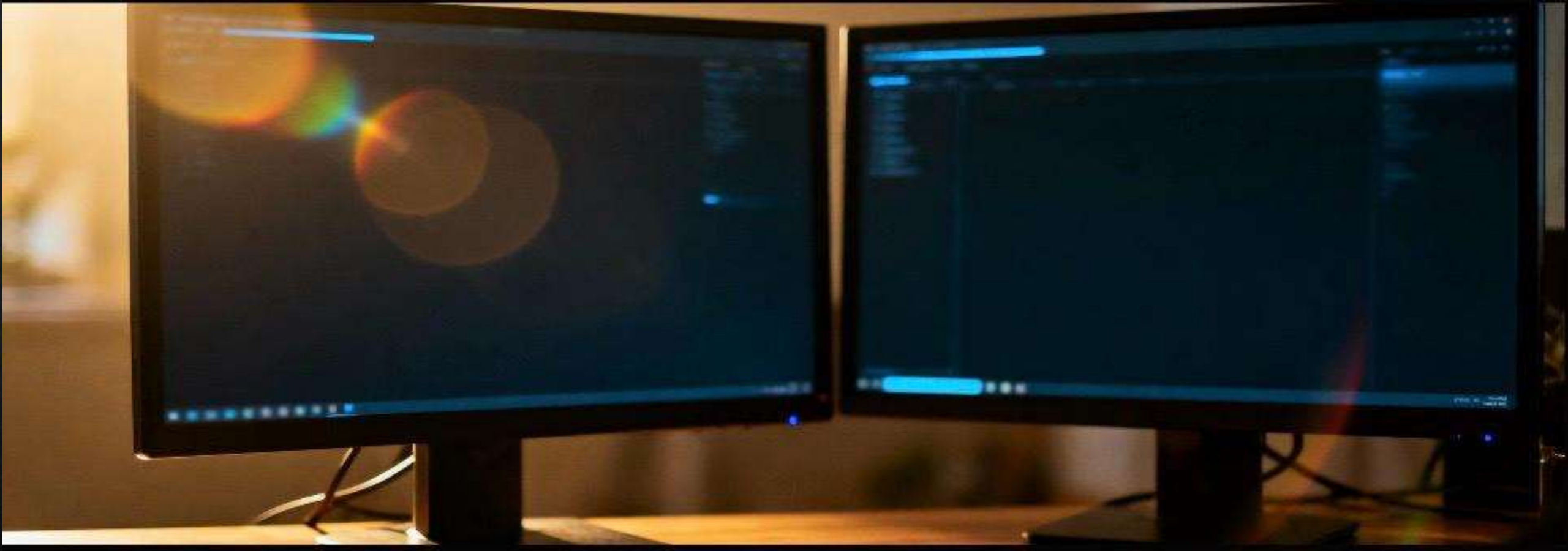
AI Tutor: It seems the problem details are missing. Could you please provide the specific question or task you'd like help with? Once you share the problem, I'll assist you further! 😊

You: hello

AI Tutor: Hello! 🙌 How can I help you today?

You: what is gdp

AI Tutor: **GDP** stands for **Gross Domestic Product**. In simple terms, it is the total monetary value of all the finished goods and services produced within a country's borders during a specific time period.



Live App: <https://ai-learning-agent-1.onrender.com>

GitHub: https://github.com/AdityaSharma-12/ai_learning_agent.git

CONCEPT NOTE

AI Tutor Pro: An AI-Powered Personalized Learning Agent for Rural Students

1. Introduction & Background

In rural regions such as Asansol and Paschim Bardhaman, students face significant educational challenges due to limited access to qualified teachers and quality learning resources. Additionally, infrastructural limitations such as low-end devices and unstable internet connectivity further hinder effective digital learning. AI Tutor Pro is proposed as an intelligent, lightweight, and multilingual AI-based learning assistant that aims to provide real-time academic support.

2. Problem Statement

- 1 Teacher Shortage: Lack of sufficient teachers limits personalized guidance.
- 2 Language Barrier: Most educational content is in English, creating difficulty for regional language students.
- 3 Hardware Constraints: Many AI tools require high-performance devices.

3. Proposed Solution

AI Tutor Pro is a full-stack web-based application leveraging Large Language Models (LLMs) to provide interactive and personalized tutoring.

- 1 Adaptive Learning: Customized explanations based on student level.
- 2 Multilingual Support: Supports Hindi and Bengali.
- 3 Cost-Effective Infrastructure: Uses OpenRouter free-tier models.
- 4 Device Compatibility: Runs on any browser.

4. Technical Architecture

- 1 Backend: Flask (Python).
- 2 Frontend: HTML, CSS, JavaScript.
- 3 API Integration: Secure handling using environment variables.
- 4 Deployment: Hosted on Render.

5. Implementation Strategy

- 1 Target Users: Rural students in Asansol region.
- 2 Value Proposition: Affordable AI tutoring.
- 3 Distribution: Government schools and NGOs.
- 4 Scalability: Expand across rural India.

6. Expected Outcomes

- 1 Improved access to education.
- 2 Better learning outcomes.
- 3 Increased digital literacy.
- 4 Scalable solution.

7. Project Information

- 1 Lead Developer: Aditya Sharma
- 2 Course: B.Tech CSE (Final Year)
- 3 Roll Number: 78
- 4 GitHub: https://github.com/AdityaSharma-12/ai_learning_agent.git
- 5 Live App: <https://ai-learning-agent-1.onrender.com>

Conclusion

AI Tutor Pro aims to bridge the rural education gap using accessible AI technology, ensuring inclusive and scalable learning solutions.